Software Requirements Specification

PlugPlan SG: arrival-aware EV charging plans for Singapore drivers

Field	Value
Version	2.0 Final
Date	10 November 2026
Course	SC2006 Software Engineering, Lab 5
Author	Team SCE3-04
AI assistance	Drafting and consistency checks used OpenAI Codex and Claude Code.

Revision history

Version	Date	Change
1.0	4 September 2026	Lab 1 requirements baseline.
1.5	25 October 2026	Lab 3 use case model refined for the design.
2.0	10 November 2026	Synchronised with the final implementation. Added FR-26 to FR-30; updated requirements are marked with an asterisk; glossary aligned with the code; traceability extended to implementation and tests.

1. Introduction

1.1 Purpose

This document specifies the final requirements of PlugPlan SG version 1.0, the system delivered by Team SCE3-04.

1.2 Document conventions

FR and NFR identify requirements, UC identifies use cases, and D and M identify desktop and mobile UI frames. BB, FS and CP identify test cases in **Lab 4/Test Cases and Results.pdf**. An asterisk marks a requirement changed since version 1.0.

1.3 Intended audience

The course supervisor and teaching assistants as the client, and developers extending the system.

1.4 Product scope

A map shows where chargers are, but a driver still has to work out whether a stop is reachable, compatible, long enough and worth the detour. PlugPlan SG answers those questions for one trip: it calculates how much charge is enough and offers a short, explained list of stops.

The system does not reserve chargers, process payment, manage queues, control charging hardware, refresh automatically in the background, or guarantee availability. Results are estimates.

1.5 References

- SC2006 Lab Manuals 1 to 5, FAQ v4.1 and SRS Template.
- LTA DataMall and OneMap API documentation.
- Companion documents: Class Diagrams.pdf and Dialog Map.pdf in this folder; Test Cases and Results.pdf in Lab 4.

2. Overall Description

2.1 Product perspective

PlugPlan SG is a new web application. A Next.js browser client calls one authenticated API route. The server combines LTA DataMall charging data and OneMap routes, runs the calculation, and persists results in Supabase with row-level security.

2.2 Product functions

- Accounts with Driver and Moderator roles, and one EV profile per Driver.
- Arrival-aware charging plans with an automatic or manual target.
- Five ranking strategies, comparison, stop selection and Google Maps navigation.
- Immutable plan versions, reconfiguration, manual refresh and fallback.
- Community issue reports with Moderator review that feeds back into planning.

2.3 User classes

User class	Description
Driver	Registers publicly, manages a vehicle profile, plans journeys and reports charger issues.
Moderator	Provisioned by the team. Reviews reports and records justified lifecycle decisions.
External Data Providers	OneMap and LTA DataMall, supporting actors for LIVE data.

2.4 Operating environment

Current desktop and mobile browsers. Deployed on Vercel (Singapore region) with Supabase for authentication, PostgreSQL and private image storage.

2.5 Constraints

- Provider calls in one request share a time budget of about 3.9 seconds.
- Provider credentials stay on the server.
- OneMap access tokens expire and must be renewed.

2.6 Assumptions and dependencies

- Charging efficiency is 90%; targets above 80% show a taper warning.
- Consumption is the Driver's declared value.
- LTA DataMall and OneMap remain available; when they fail, the system labels degraded results or reports the failure.
- Google and GitHub sign-in are not part of the delivered baseline.

3. Functional Requirements

Accounts and profile

ID	Priority	Requirement
FR-01*	High	The system shall create a Driver account when a user submits a unique email address and a password of at least 10 characters containing a letter and a number.
FR-02	High	The system shall authenticate a registered user and establish a session containing the user's Driver or Moderator role.
FR-03*	High	The system shall allow a Driver to create, view, update, and delete one EV profile containing vehicle name, usable battery capacity, energy consumption, supported plug types, AC/DC charging limits, reserve SOC, default detour, minimum charger power, and maximum walking distance.
FR-04	High	The system shall reject an EV profile containing a missing or out-of-range required value and display a field-specific validation message.

Charging plan creation

ID	Priority	Requirement
FR-05*	High	The system shall create a charging-plan request from the saved EV profile, origin, destination, current SOC, departure time, maximum charging duration, maximum detour, ranking strategy, and optional manual target SOC.
FR-06*	High	The system shall use OneMap to search places for Driver selection and to obtain route distance and duration, or report that a route is unavailable.
FR-07	High	The system shall retrieve the latest usable charging-station and connector snapshot through the LTA DataMall backend adapter.
FR-08	High	The system shall estimate the arrival SOC at each candidate station from route distance, current SOC, usable battery capacity, and declared vehicle consumption.
FR-09	High	The system shall exclude a candidate station when the estimated battery energy on arrival is below zero.
FR-10	High	The system shall exclude a candidate station when it has no connector matching the vehicle's supported plug types.
FR-11*	High	The system shall exclude a candidate station when it is closed during the planned charging window. Missing or unparseable hours are assumed open and disclosed with a warning.
FR-12	High	The system shall exclude a candidate station when its route detour exceeds the Driver's maximum detour.
FR-13	High	The system shall calculate the automatic target SOC required to travel from the charger to the destination while retaining the configured reserve SOC.
FR-14	High	The system shall allow the Driver to replace the automatic target with a valid manual target while continuing to display the automatic target for comparison.
FR-15	High	The system shall estimate required charging energy and charging time using the lower of connector power and the applicable vehicle AC/DC charging limit.
FR-16	High	The system shall classify a candidate as dwell-infeasible when its estimated charging time exceeds the submitted maximum charging duration.
FR-17	High	The system shall estimate charging cost for supported per-kWh or per-hour prices and display 'Price unavailable' for a missing or unsupported price.
FR-27*	High	The system shall exclude a candidate station when its walking distance exceeds the Driver's limit or cannot be verified against a set limit, or when no connector reaches the minimum charger power.
FR-28*	High	After ranking and before forming the shortlist, the system shall exclude a station supported by an active, Moderator-verified suspected-fault report.
FR-29*	High	The system shall provide an explicitly selected and labelled Demo Fixture mode and shall never silently substitute demo data for live data.

Recommendations

ID	Priority	Requirement
FR-18	High	The system shall rank feasible candidates using the selected Fastest overall journey, Cheapest charging, Availability-first, Minimum detour, or Balanced preset.
FR-19*	High	The system shall display up to three shortlisted recommendations with arrival SOC, target SOC, required energy, charging time, estimated cost, ranking explanation, source timestamps, and data-state label, and shall explain each excluded station or a no-option result.
FR-20	Medium	The system shall allow the Driver to compare two or three shortlisted recommendations side by side.
FR-26*	High	The system shall allow the Driver to select one recommendation from the current Plan Version without creating a new version or making a reservation.
FR-30*	Medium	The system shall open Google Maps driving directions from the origin to the destination through a chosen charging stop.

Plan lifecycle

ID	Priority	Requirement
FR-21	High	The system shall recalculate a plan after the Driver changes an input or strategy and store the result as a new immutable Plan Version.
FR-22	High	The system shall allow the Driver to manually refresh a saved plan using the latest usable external-data snapshot.
FR-23*	High	When a refreshed plan makes the previously selected recommendation unavailable, the system shall clear the selection and identify the next-ranked alternative for explicit Driver selection, or state that no feasible fallback exists.

Community reports

ID	Priority	Requirement
FR-24*	Medium	The system shall allow a Driver to submit a report that expires after 24 hours by default, containing a station or connector, category, note, and optional JPEG, PNG or WebP image up to 5 MB, limited to five reports per hour, with duplicates answered by a safe summary.
FR-25	Medium	The system shall allow only a Moderator to verify, reject, merge, resolve, or expire an issue report while recording the decision reason and timestamp.

4. External Interface Requirements

4.1 User interfaces

The UI frames and transitions are specified in Dialog Map.pdf. The design mockups and the implemented screens are in Appendix B.

4.2 Software interfaces

Interface	Use	Requirement
OneMap	Place search, driving routes, map tiles	FR-06
LTA DataMall	Charging station and connector status	FR-07
Supabase	Authentication, PostgreSQL with row-level security, private image storage	FR-01 to FR-05, FR-21 to FR-26
Google Maps	Outbound directions link only	FR-30

4.3 Communications interfaces

The browser sends JSON to `POST /api/app` over HTTPS with a bearer token. The `action` field selects one of 14 operations. Errors are JSON with safe messages and, where relevant, field messages.

5. Non-Functional Requirements

Evidence dates from the 15 October 2026 verification record unless stated. Since that record, the application has changed only by adding the Google Maps link, pinning the deployment region and adding code comments. Measurements for NFR-01 were taken on the team's reference environment.

ID	Quality	Requirement	Evidence	Status
NFR-01	Performance	With cached provider data, 95% of 100 recommendation requests under 20 concurrent users shall complete within 3.0 seconds on the team's documented reference environment.	100/100 requests succeeded; p95 1,918.67 ms.	Met
NFR-02	Failure response	When an external provider times out or returns an error, the system shall display a degraded-state result or explicit failure within 5 seconds.	Controlled stalled LIVE create timed out in about 3.98 s; `api.timeout.test.ts`.	Met
NFR-03	Freshness	The system shall label LTA data older than 10 minutes as Stale and shall display observedAt and retrievedAt wherever a recommendation is shown.	Boundary tests BB-66 (9:59) and BB-67 (10:00).	Met
NFR-04	Calculation correctness	Approved fixtures shall produce 100% correct feasibility decisions and numerical results within +/-0.1 SOC percentage point, +/-0.1 kWh, +/-1 minute, and +/-S$0.01.	All fixture tests pass (BB-03 and engine suite); fixtures approved by the team.	Met
NFR-05	Security	All authenticated traffic shall use TLS 1.2 or later; passwords shall be stored only as salted one-way hashes; no provider secret shall appear in client code, repository history, or application logs.	TLS 1.2 handshake to the deployment confirmed 10 Nov 2026. Hash format and secret scan recorded 15 Oct 2026.	Met
NFR-06	Authorization	Every Moderator endpoint shall reject an authenticated Driver with HTTP 403, as verified by an automated test for each protected operation.	Driver received 403 for Verify, Reject, Merge, Resolve and Expire.	Met
NFR-07	Privacy	After a Driver deletes an EV profile or saved plan, it shall become inaccessible through the application within 60 seconds; unsaved origin and destination inputs shall not be retained.	Deletion and confirmed absence in API journeys; E2E: unsaved locations cleared on reload.	Met

ID	Quality	Requirement	Evidence	Status
NFR-08	Accessibility	The nine core frames shall have zero serious or critical automated accessibility violations, and the implemented core journeys shall support keyboard-only completion.	Zero serious or critical axe violations in all nine frames; keyboard E2E passed.	Met
NFR-09	Usability	At least 4 of 5 first-time representative users shall create a plan and compare two recommendations without assistance within 3 minutes.	First-use study conducted by the team.	Met
NFR-10	Reliability and testability	The deterministic fixture-based recommendation journey shall complete successfully in 100 consecutive runs, and energy, feasibility, cost, and ranking modules shall achieve at least 80% branch coverage.	100/100 API journeys; module branch coverage 100%, 86.47%, 100%, 85.33%.	Met

6. Traceability

From requirement to use case, UI frame, design class, implementation and test. Paths are under **PlugPlan SG/src** and **PlugPlan SG/tests**. E2E refers to `tests/e2e/app.spec.ts`.

Req.	Use case	Frame	Design class	Implementation	Tests
FR-01	UC-01	D-00	AuthScreen, SupabaseClient	`plugplan-app.tsx`; Supabase Auth	E2E email registration
FR-02	UC-01	D-00	AuthScreen, AuthGuard	`supabase.ts` requireUser	E2E D-00; `api.request.test.ts`
FR-03	UC-02	D-01	PlugPlanApp, ProfileRepository	`db.ts` saveProfile, deleteProfile	E2E D-01 to D-05
FR-04	UC-02	D-01	ValidationSchemas	`validation.ts` profileSchema	`domain.validation.test.ts`; BB-29 to 36
FR-05	UC-03	D-02	PlacePicker, ServiceHandler, ValidationSchemas	`service.ts` plan.create; planInputSchema	BB-01, 05, 08, 13, 14, 19, 61, 64, 65
FR-06	UC-03	D-02	PlacePicker, ProviderGateway	`providers.ts` searchPlaces, getRoutes	`providers.normalization.test.ts`; FS-02
FR-07	UC-03, UC-06	D-03	ProviderGateway, ProviderCache	`providers.ts` getLiveSnapshot	`providers.normalization.test.ts`
FR-08	UC-03	D-03	PlanEngine	`engine.ts` failureForStation	BB-02, 03
FR-09	UC-03	D-03	PlanEngine	failureForStation	BB-06, 07; FS-03
FR-10	UC-03	D-03	PlanEngine	failureForStation, canonicalPlugType	BB-41, 42; FS-05
FR-11	UC-03	D-03	PlanEngine	parseOpeningHours, coversChargingWindow	BB-43 to 46; FS-17, 18, 24
FR-12	UC-03	D-02, D-03	PlanEngine	failureForStation	BB-09 to 12; FS-04
FR-13	UC-03	D-03	PlanEngine	failureForStation	BB-20, 30, 31, 34, 37 to 40; FS-09, 11
FR-14	UC-03, UC-05	D-02, D-03	PlanEngine	failureForStation	BB-20 to 28; FS-10, 22
FR-15	UC-03	D-03	PlanEngine	failureForStation	BB-47 to 50; FS-13
FR-16	UC-03	D-03	PlanEngine	failureForStation	BB-15 to 17; FS-16, 25
FR-17	UC-03, UC-04	D-03, D-04	PlanEngine	estimateCost	BB-51 to 55; FS-19 to 21
FR-18	UC-03 to UC-05	D-03	PlanEngine	assignScores, compareCandidates	BB-56 to 61
FR-19	UC-03, UC-04	D-03, M-02	PlanEngine, PlugPlanApp	calculatePlan; results view	BB-62, 63; CP-01, 11, 12; E2E D-01 to D-05
FR-20	UC-04	D-04	PlugPlanApp	Compare view	E2E D-01 to D-05; E2E keyboard
FR-21	UC-05	D-02, D-03	ServiceHandler, PlanRepository	plan.reconfigure; savePlanVersion	E2E D-01 to D-05
FR-22	UC-06	D-03	ServiceHandler, ProviderGateway	plan.refresh	`api.refresh-selection.test.ts`
FR-23	UC-06	D-03	ServiceHandler	explainInvalidatedSelection	`api.refresh-selection.test.ts`
FR-24	UC-07	D-05	PlugPlanApp, ReportRepository	report.create; submitReport	E2E D-01 to D-05; `domain.validation.test.ts`
FR-25	UC-08	D-06	AuthGuard, ReportRepository	report.moderate; moderateReport	E2E D-06
FR-26	UC-04	D-03, D-04	PlanRepository	plan.select; selectRecommendation	E2E D-01 to D-05
FR-27	UC-03	D-03	PlanEngine	failureForStation	BB-49, 50; FS-06, 07, 15
FR-28	UC-03, UC-05, UC-06, UC-08	D-03	PlanEngine, ReportRepository	calculatePlan; getActiveFaults	BB-68, 69; CP-15, 16
FR-29	UC-03, UC-06	D-02	DemoFixtures, ServiceHandler	`fixtures.ts` createDemoScenario	CP-09; `domain.engine.test.ts`
FR-30	UC-04	D-03	MapUtilities	`map.ts` googleMapsDirectionsUrl	`map.test.ts`

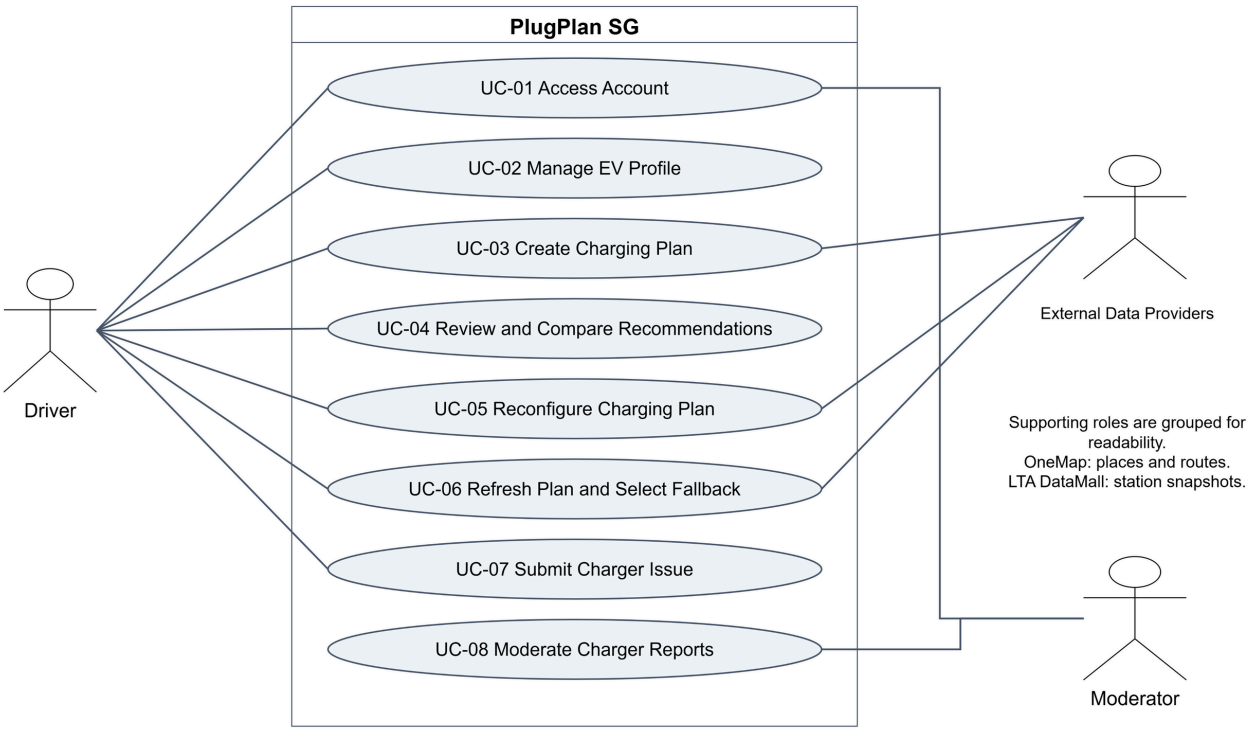
Appendix A: Glossary and Data Dictionary

Term	Definition	Attributes and constraints	Relationships
User (AppUser)	A person who accesses PlugPlan SG.	id; unique email; role {DRIVER, MODERATOR}; status	A Driver owns one EV Profile, Charging Plans and Issue Reports. A Moderator records Moderation Decisions.
EV Profile (VehicleProfile)	Planning assumptions for one Driver's vehicle.	name; capacityKWh 10-250; consumptionKWhPer100Km 5-50; plugTypes (one or more); maxAcKw 1-50; maxDcKw 1-400; reserveSoc 5-40 (default 20); defaultDetourMinutes 0-120; minChargerKw 0-400; maxWalkingM >= 0 (0 means no limit)	At most one per Driver. Copied into each Plan Version.
Charging Plan (SavedPlan)	A Driver's persistent planning record.	id; title; createdAt; currentVersionId; selectedRecommendationId	Owns one or more Plan Versions.
Plan Version	An immutable result of one calculation.	id; planId; number; createdAt; reason {CREATE, RECONFIGURE, REFRESH}; result (PlanResult)	Belongs to one Charging Plan.
Plan Input	The Driver's request.	origin; destination; currentSoc 0-100; departureAt; maxChargingMinutes 5-720; maxDetourMinutes 0-120; strategy; manualTargetSoc null or 0-100	Stored in each result.
Charging Station	A physical EV charging location.	id; name; address; postal; lat; lng; operator; openingHours; openingHoursAssumed	Contains Connectors and may have Issue Reports.
Connector	One charging interface at a station.	id; plugType; powerType {AC, DC}; powerKw; price; priceType {kWh, h, free, unknown}; status {AVAILABLE, OCCUPIED, UNAVAILABLE}	Belongs to one Charging Station.
Official Snapshot	Timestamped provider data used by a calculation.	source; state; observedAt; retrievedAt; stations	Its source and times are stored in each result.
Data State	Provenance label of the data.	LIVE; CACHED; STALE; DEMO_FIXTURE	Stale from 10 minutes of age; Demo Fixture is explicitly selected.
Route Estimate	A OneMap route result.	distanceKm; durationMinutes; optional geometry	Used for energy and detour calculation.
Candidate Route	The legs via one station.	stationId; toStation; fromStation; optional walkingMeters	One per assessed station.
Recommendation	A ranked assessment of one connector.	stationId; connectorId; arrivalSoc; automaticTargetSoc; targetSoc; requiredBatteryKWh; requiredChargerKWh; chargingMinutes; estimatedCost or null; detourMinutes; totalJourneyMinutes; availability; score; rank; explanations; warnings	At most three per Plan Version, one per station.
Exclusion	A coded reason a station is not offered.	stationId; stationName; code; reason	Examples: UNREACHABLE, DETOUR_LIMIT, DWELL_LIMIT, ACTIVE_FAULT_MARK.
Ranking Strategy	A fixed preset.	BALANCED (time 0.35, cost 0.25, availability 0.25, detour 0.15); FASTEST; CHEAPEST; AVAILABILITY; DETOUR	Selected per Plan Version.
Issue Report	A time-limited community report.	id; userId; stationId; connectorId or null; category; status; note; imagePath; expiresAt; revision; mergedIntoId	Receives zero or more Moderation Decisions.
Report Status	Lifecycle state.	PENDING; VERIFIED; REJECTED; MERGED; RESOLVED; EXPIRED	Constrains valid transitions.
Moderation Decision	An auditable Moderator action.	id; action {VERIFY, REJECT, MERGE, RESOLVE, EXPIRE}; reason; moderatorId; createdAt; resultingStatus	Appended to one Issue Report.
Fault Evidence	A verified, unexpired fault mark applied in planning.	stationId; reportId; decisionId; expiresAt	Excludes the station after ranking.

Appendix B: Analysis Models

B.1 Use Case Model

Complete use case diagram



Solid lines represent participation. Authentication is a precondition; no include or extend relationship is asserted.

Diagram conventions

- The system boundary contains behaviour owned by PlugPlan SG.
- Solid actor associations indicate participation, not execution order.
- OneMap and LTA DataMall are grouped as **External Data Providers** in the diagram for readability. OneMap supplies place and route data; LTA DataMall supplies station and connector snapshots.
- No `<<include>>` or `<<extend>>` relationship is asserted. Authentication is a precondition for UC-02 to UC-08, not a sub-use-case re-executed within each use case.
- Supabase is an internal implementation component for authentication and persistence, so it is not shown as an external domain actor.

Use case catalogue

ID	Use case	Primary actor	Supporting actor
UC-01	Access Account	Driver or Moderator	-
UC-02	Manage EV Profile	Driver	-
UC-03	Create Charging Plan	Driver	OneMap and LTA DataMall, when LIVE data is used
UC-04	Review and Compare Recommendations	Driver	-
UC-05	Reconfigure Charging Plan	Driver	OneMap and LTA DataMall, when fresh data is required
UC-06	Refresh Plan and Select Fallback	Driver	OneMap and LTA DataMall, when LIVE data is used
UC-07	Submit Charger Issue	Driver	-
UC-08	Moderate Charger Reports	Moderator	-

UC-01 - Access Account

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	24 October 2026

Field	Value
Actor(s)	Driver or Moderator
Priority	High
Frequency	At the start of each authenticated session
Traceability	FR-01, FR-02; D-00

Description. A new Driver registers, or an existing Driver or Moderator signs in and reaches the workspace allowed by the account role.

Preconditions

- The user can access the PlugPlan SG web application.
- For sign-in, an active account exists.

Postconditions

- An authenticated session is established and the application has loaded the user's persisted role.
- The user is shown the Driver or Moderator workspace; after logout, the session is invalidated.

Normal flow of events

1. The user selects the sign-in function.
2. The system displays the email and password form.
3. The user enters credentials and submits the form.
4. The system validates the credentials and loads the user's role.
5. The system establishes the session and displays the role-appropriate workspace.

Alternative flows

- UC-01.A1 - At step 1, a new user selects registration, supplies a unique email and compliant password, and the system creates a Driver account before continuing at step 5.
- UC-01.A2 - From an authenticated workspace, the user selects logout; the system invalidates the session and returns to the access screen.

Exceptions

- UC-01.E1 - Invalid credentials or an inactive account cause access to be denied with a generic error that does not disclose whether the email exists.
- UC-01.E2 - If the authentication service is unavailable, no session is created and the system displays a retry instruction.

Includes: None.

Special requirements: NFR-05 and NFR-06 apply. Passwords and session tokens must never be logged or displayed.

Assumptions: Moderator accounts are provisioned by the team; public registration creates Driver accounts only. Google and GitHub OAuth are not part of the submitted baseline.

Notes and issues: Authentication is modelled as its own user goal. It is a precondition, not an `<<include>>` relationship, for UC-02 to UC-08.

UC-02 - Manage EV Profile

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Driver
Priority	High
Frequency	Occasionally; before first plan and when vehicle assumptions change
Traceability	FR-03, FR-04; D-01

Description. The Driver creates, updates or deletes the validated EV and planning assumptions used by charging calculations.

Preconditions

- The Driver is authenticated.
- The Driver workspace is available.

Postconditions

- On save, exactly one validated EV Profile is stored for the Driver.
- On confirmed deletion, the Driver has no active EV Profile; existing saved Plan Versions remain historical records.

Normal flow of events

1. The Driver opens EV Profile.
2. The system displays the current profile, or a setup form with default values if none exists.
3. The Driver enters vehicle, battery, connector, charging and route-preference values.
4. The Driver submits the profile.

5. The system validates all required fields and allowed ranges.
6. The system creates or updates the profile and confirms the save.

Alternative flows

- UC-02.A1 - At step 3, the Driver cancels; the system discards unsaved edits and restores the last saved values.
- UC-02.A2 - At step 3, the Driver selects Delete; after confirmation, the system removes the profile and displays the empty state.

Exceptions

- UC-02.E1 - Missing, inconsistent or out-of-range values are rejected with field-specific messages; the Driver returns to step 3.
- UC-02.E2 - If persistence fails, no partial update is committed and the entered values remain available for retry.

Includes: None.

Special requirements: Capacity, consumption, plug type, AC/DC power, reserve, detour, minimum charger power and maximum walking distance must be validated server-side.

Assumptions: The MVP permits at most one EV Profile per Driver. A valid profile is required before planning. A new profile defaults to 20% reserve SOC; the Driver can configure it from 5% to 40%.

Notes and issues: Changing the profile affects newly created or reconfigured versions; it does not rewrite immutable historical Plan Versions.

UC-03 - Create Charging Plan

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Driver (primary); External Data Providers - OneMap and LTA DataMall (supporting when LIVE data is used)
Priority	Critical
Frequency	Once or more per planned journey
Traceability	FR-05 to FR-19, FR-27 to FR-29; D-02, D-03

Description. The Driver submits journey constraints and receives a saved Plan Version containing up to three feasible, ranked and explained charging recommendations.

Preconditions

- The Driver is authenticated and has a valid EV Profile.
- Origin and destination can be selected through OneMap search, or Demo Fixture mode is explicitly selected.

Postconditions

- A Charging Plan and immutable Plan Version are stored with inputs, data-state evidence and calculation results.
- Up to three feasible recommendations are displayed, or a clear no-feasible-option result is shown.

Normal flow of events

1. The Driver enters origin, destination, current state of charge, departure time, maximum charging time, maximum detour, ranking strategy and optional manual target SOC, then submits.
2. The system validates the inputs and resolves the selected places through OneMap.
3. The system obtains a station snapshot from LTA DataMall and direct/via route results from OneMap.
4. The system computes compatibility, reachability, arrival charge, target energy, charging time, detour, dwell and available cost evidence.
5. The system excludes infeasible candidates and ranks eligible station candidates using the selected strategy. After ranking, it excludes stations supported by at least one active, Moderator-verified `SUSPECTED_FAULTY_CONNECTOR` report, including reports attached to a connector.
6. The system takes up to the three highest-ranked surviving stations, stores the Plan Version and displays the recommendations with assumptions, source timestamps and data-state labels.

Alternative flows

- UC-03.A1 - At step 1, the Driver explicitly selects Demo Fixture mode; the system uses labelled fixture inputs and continues at step 4 without calling live providers.
- UC-03.A2 - At step 2, a place search returns multiple matches; the Driver selects one result and the system continues at step 3.
- UC-03.A3 - At step 4, a manual target is supplied; the system validates and uses it instead of the calculated target.
- UC-03.A4 - At step 5, no charging stop is required or no feasible connector remains; the system stores and explains that result instead of presenting three options.

Exceptions

- UC-03.E1 - If no valid route or usable station snapshot is available, the use case fails, no partial Plan Version is stored and the Driver's inputs remain available.
- UC-03.E2 - A provider or calculation error is displayed with its actual LIVE, CACHED, STALE or DEMO_FIXTURE state; the system never silently substitutes demo data.

Includes: None.

Special requirements: Inputs and computed outputs must be validated server-side. Each saved result retains its inputs, copied profile assumptions, source timestamps, route estimates and recommendation metrics for later review.

Assumptions: A new EV Profile defaults to 20% reserve SOC, configurable by the Driver from 5% to 40%. Charging efficiency is 90%, targets above 80% display a taper warning, and price may be unavailable.

Notes and issues: OneMap supplies search and route data; LTA DataMall supplies station/connector snapshots when LIVE data is used. Fault exclusion is station-level and occurs after ranking but before the three-option shortlist is formed. A connector-target report supports exclusion of its parent station only while it is categorized `SUSPECTED_FAULTY_CONNECTOR`, verified and unexpired.

UC-04 - Review and Compare Recommendations

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	10 November 2026
Actor(s)	Driver
Priority	High
Frequency	After plan creation, reconfiguration or refresh
Traceability	FR-18 to FR-20, FR-26, FR-30; D-03, D-04

Description. The Driver reviews and compares recommendations, then may select a preferred recommendation from the current Plan Version without making a reservation.

Preconditions

- The Driver is authenticated.
- The Driver owns a Charging Plan with at least one Plan Version.
- To select a recommendation, the current Plan Version must contain that recommendation; historical versions are read-only for selection.

Postconditions

- The Charging Plan stores the selected recommendation ID as a mutable plan-level selection, which must refer to a recommendation in its current Plan Version.
- Changing the selection does not create a new Plan Version.
- No station reservation, payment or queue position is created.

Normal flow of events

1. The Driver opens a saved plan and selects a Plan Version to review.
2. The system displays its ranked cards, route context, connector details, charging and cost estimates, detour, battery and energy estimates, availability, explanations, assumptions, source and data-state labels, and timestamps.
3. The Driver selects two or three displayed options for comparison, or proceeds with the default comparison of all displayed recommendations.
4. The system displays the selected options in a consistent metric order.
5. The Driver reviews the trade-offs and, when viewing the current Plan Version, selects a preferred recommendation.
6. The system updates the Charging Plan's selection and confirms it without creating a Plan Version.

Alternative flows

- UC-04.A1 - At step 3, fewer than two recommendations exist; comparison is disabled, while any available result and exclusions remain reviewable.
- UC-04.A2 - At step 4, price data is missing; the system displays Price unavailable and does not invent a price.
- UC-04.A3 - At step 1, the Driver opens a historical Plan Version; it can be reviewed and compared, but recommendation selection is disabled.
- UC-04.A4 - At step 5, the Driver chooses a different recommendation in the current Plan Version; the system replaces the plan-level selection without creating a new version.
- UC-04.A5 - At step 5, the Driver selects Navigate via this stop; the system opens Google Maps driving directions from the origin to the destination through that charging stop in a new tab. Plan state does not change and no reservation is made.

Exceptions

- UC-04.E1 - If the plan is missing or not owned by the Driver, access is denied and no selection is changed.
- UC-04.E2 - If saving the selection fails, the previous selection remains unchanged and a retry message is shown.

Includes: None.

Special requirements: All compared options must use the same units and calculation assumptions. A maximum of three options may be compared at once.

Assumptions: Recommendations are decision support, not guaranteed charger availability or a booking.

Notes and issues: Saved-plan browsing and Plan Version switching are supporting behaviours of review, not additional goal-level use cases.

UC-05 - Reconfigure Charging Plan

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Driver (primary); External Data Providers - OneMap and LTA DataMall (supporting when LIVE data is required)
Priority	High
Frequency	Whenever journey constraints or ranking strategy change
Traceability	FR-05, FR-14, FR-18, FR-21, FR-27, FR-28; D-02, D-03

Description. The Driver changes plan inputs or strategy and receives a new immutable Plan Version while retaining the previous version.

Preconditions

- The Driver is authenticated and owns the Charging Plan.
- At least one Plan Version already exists.

Postconditions

- A new immutable Plan Version is stored and displayed when recalculation succeeds.
- The plan-level selection is retained only if the same recommendation remains in the new current-version shortlist; otherwise it is cleared.
- The previous version remains available for comparison and history.

Normal flow of events

1. The Driver opens a plan and chooses to edit inputs or change strategy.
2. The system loads the current version's values into the reconfiguration view.
3. The Driver changes one or more values and submits.
4. The system validates the changes and either reuses sufficiently recent evidence or obtains required OneMap/LTA data.
5. The system recalculates feasibility and charging estimates, ranks eligible stations, then excludes stations supported by active, Moderator-verified `SUSPECTED_FAULTY_CONNECTOR` reports before taking up to three survivors.
6. The system stores and displays a new Plan Version without overwriting the previous version.

Alternative flows

- UC-05.A1 - At step 1, the Driver selects a strategy tab; the system treats this as a strategy-only reconfiguration and continues at step 4.
- UC-05.A2 - At step 3, the Driver cancels; the system discards unsaved edits and returns to the current version.
- UC-05.A3 - At step 5, a formerly recommended connector becomes infeasible; it is omitted from the new ranked result and the explanation identifies the controlling constraint.

Exceptions

- UC-05.E1 - If required route or station data cannot be obtained or reused, no partial version is stored and the previous version remains visible.
- UC-05.E2 - If the Driver attempts to leave with unsaved edits, the system requests confirmation before discarding them.

Includes: None.

Special requirements: Version creation must be atomic. Every version must retain its own inputs, assumptions, evidence timestamps and data-state label.

Assumptions: Reconfiguration is user-initiated; continuous background recalculation is outside scope.

Notes and issues: Provider participation is conditional because sufficiently recent cached evidence may be reused.

UC-06 - Refresh Plan and Select Fallback

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Driver (primary); External Data Providers - OneMap and LTA DataMall (supporting when LIVE data is used)
Priority	High
Frequency	Before departure or when the Driver suspects conditions changed
Traceability	FR-07 to FR-19, FR-22, FR-23, FR-27 to FR-29; D-03, M-02

Description. The Driver manually requests newer route and station evidence and selects a fallback if the previously chosen connector is no longer feasible.

Preconditions

- The Driver is authenticated and owns a Charging Plan with a current Plan Version.
- The plan has enough stored inputs to be recalculated.

Postconditions

- On success, a new Plan Version is stored with refreshed evidence and feasibility results.
- The plan-level selection is retained only if the same recommendation remains in the refreshed shortlist; otherwise it is cleared and the next-ranked surviving option, if any, is identified for explicit selection.

Normal flow of events

1. The Driver selects Refresh on the current Plan Version.
2. The system requests the latest usable OneMap route data and LTA station snapshot.
3. The system recalculates feasibility and estimates, ranks eligible stations, then excludes stations supported by active, Moderator-verified `SUSPECTED_FAULTY_CONNECTOR` reports before taking up to three survivors.
4. The system compares the previous selection with the refreshed result.
5. The system stores and displays a new Plan Version, retaining the plan-level selection only if the same recommendation remains in the new shortlist; otherwise it clears that selection.
6. The Driver reviews the evidence, keeps any retained selection or explicitly selects a fallback recommendation.

Alternative flows

- UC-06.A1 - At step 2, only cached or stale evidence is usable; the system labels its state and age before continuing at step 3.
- UC-06.A2 - At step 4, the selected recommendation remains in the refreshed shortlist; it remains selected for the new current version.
- UC-06.A3 - At step 4, the selected recommendation is no longer in the refreshed shortlist; the system clears the plan-level selection, explains why and identifies the highest-ranked surviving option for step 6.
- UC-06.A4 - In Demo Fixture mode, the system refreshes from the labelled fixture snapshot rather than live providers.

Exceptions

- UC-06.E1 - If no usable route or station snapshot exists, the refresh fails, no new version is stored and the previous version remains unchanged.
- UC-06.E2 - If no feasible fallback remains, the new version states this explicitly and no recommendation is auto-selected.

Includes: None.

Special requirements: Refresh must never silently replace live data with demo data. A fallback requires explicit Driver selection.

Assumptions: The latest usable snapshot may be LIVE, CACHED, STALE or explicitly DEMO_FIXTURE.

Notes and issues: Refresh is manual in the MVP; there is no continuous monitoring or guaranteed real-time charger availability.

UC-07 - Submit Charger Issue

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Driver
Priority	Medium
Frequency	When the Driver observes a charger issue
Traceability	FR-24; D-05

Description. The Driver submits a categorized, expiring community report with an optional photo for a station or connector.

Preconditions

- The Driver is authenticated.
- A station or connector context is available from a recommendation or saved plan.

Postconditions

- A valid non-duplicate report is stored as pending with a default 24-hour expiry and appears in the Driver's report history.
- A pending submission alone does not exclude the station from recommendations.

Normal flow of events

1. The Driver opens Report Issue for a station or connector.
2. The system prefills the charger context and displays category, note and optional image fields.
3. The Driver enters the evidence and submits the report.
4. The system validates the content, image type/size and submission rate.
5. The system checks for a recent matching active report.
6. The system stores the pending report and displays its reference, status and expiry.

Alternative flows

- UC-07.A1 - At step 3, the Driver submits without an image; the system continues at step 4 because an image is optional.

- UC-07.A2 - At step 5, a recent matching report exists; the system displays a safe summary and the Driver returns to the recommendation without creating a duplicate.
- UC-07.A3 - At step 2, the Driver selects operator support; the system opens the support link and no community report is created.

Exceptions

- UC-07.E1 - A non-JPEG/PNG/WebP image or an image above 5 MB is rejected without discarding the other entered fields.
- UC-07.E2 - If the limit of five reports per hour is exceeded, no report is stored and the system explains when submission can be retried.
- UC-07.E3 - If upload or persistence fails, no partial report is created and the Driver can retry.

Includes: None.

Special requirements: Notes and media must be access-controlled. Duplicate summaries must not reveal another Driver's identity or private content.

Assumptions: Default expiry is 24 hours unless a Moderator changes the lifecycle earlier.

Notes and issues: Only active, verified, unexpired `SUSPECTED_FAULTY_CONNECTOR` reports support station-level exclusion during planning; a connector-target report supports its parent station.

UC-08 - Moderate Charger Reports

Field	Value
Created by	Team SCE3-04
Last updated by	Team SCE3-04
Date created	4 September 2026
Date updated	25 October 2026
Actor(s)	Moderator
Priority	High
Frequency	Several times per day while reports are pending
Traceability	FR-02, FR-25, FR-28; D-06

Description. A Moderator reviews community evidence and records a justified lifecycle decision that may affect recommendation filtering.

Preconditions

- The Moderator is authenticated with an active Moderator role.
- At least one report exists in the moderation queue or history.

Postconditions

- The accepted lifecycle action and nonblank reason are stored with an audit entry.
- Recommendation filtering changes only when a resulting verified, unexpired `SUSPECTED_FAULTY_CONNECTOR` report supports a station-level mark.

Normal flow of events

1. The Moderator opens the report queue and selects a report.
2. The system displays report evidence, related reports, official status and audit history.
3. The Moderator selects Verify, Reject, Merge, Resolve or Expire, supplies a reason and any required merge target.
4. The system requests confirmation when required and validates role, report revision and lifecycle rules.
5. The system applies the action, records the decision and appends the audit history.
6. The system refreshes the queue and displays the resulting status and filtering effect.

Alternative flows

- UC-08.A1 - At step 3, the Moderator closes or cancels the decision form; the report keeps its existing status and no decision is recorded.
- UC-08.A2 - At step 3, Merge is selected; the Moderator chooses a valid target and the system preserves one authoritative report before continuing at step 4.
- UC-08.A3 - At step 2, official status or evidence contradicts the report; the Moderator selects Reject or Resolve and records the reason.
- UC-08.A4 - At step 3, a reason or valid merge target is missing; the system displays validation feedback and returns to step 3.

Exceptions

- UC-08.E1 - If another Moderator changed the report, the server rejects the stale revision; no decision is applied and the Moderator must refresh.
- UC-08.E2 - If the account lacks the Moderator role, access is denied and no report data or action is exposed.
- UC-08.E3 - If persistence fails, the lifecycle state and audit history remain unchanged.

Includes: None.

Special requirements: Every action requires a reason and server-side authorization; destructive or terminal actions require confirmation. Concurrency control must reject stale revisions.

Assumptions: There is no appeal workflow in the MVP. Pending, rejected, resolved and expired reports do not cause exclusion.

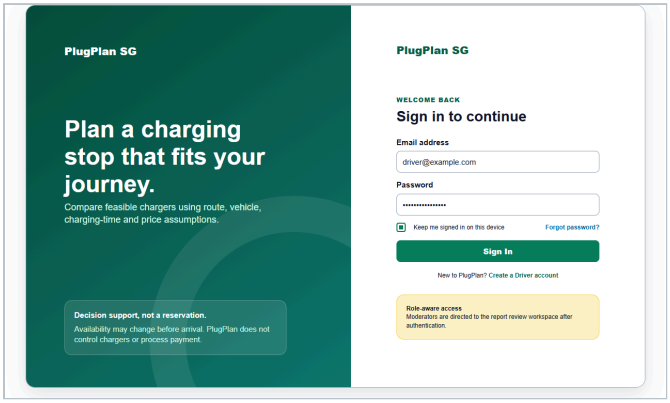
Notes and issues: A station remains marked only while at least one `SUSPECTED_FAULTY_CONNECTOR` report is VERIFIED and unexpired; a connector-target report supports its parent station. Pending, rejected, resolved, expired and non-fault reports do not cause exclusion, and merge must not create duplicate support.

Scope exclusions

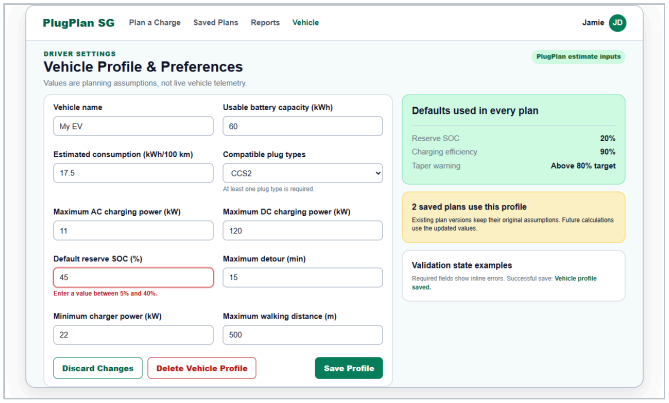
The current MVP does not perform charger reservation, payment, charger queue management, guaranteed real-time availability, automatic background refresh, or automatic fallback selection.

B.2 UI Mockups

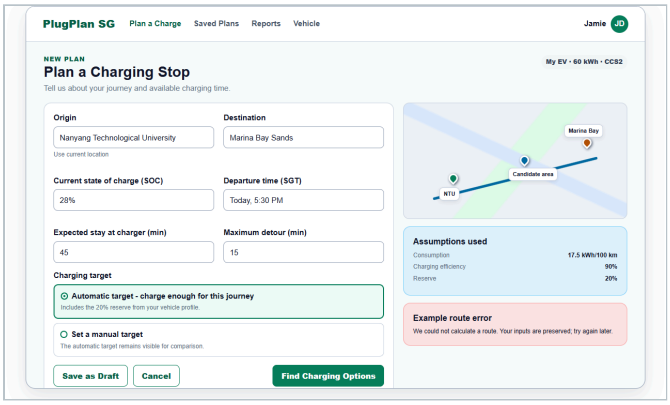
The Lab 1 design frames, which define the screen identifiers used throughout this document.



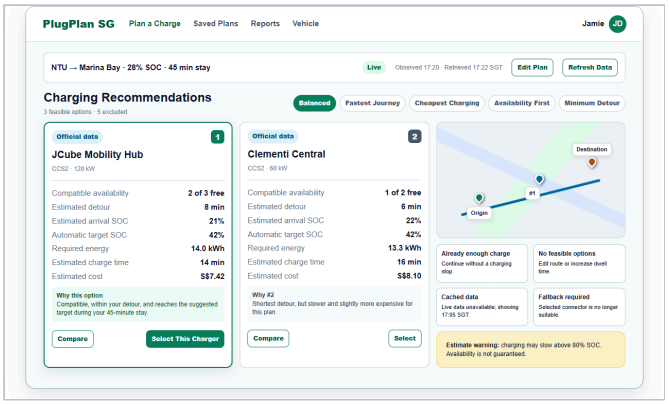
D-00 Account Access



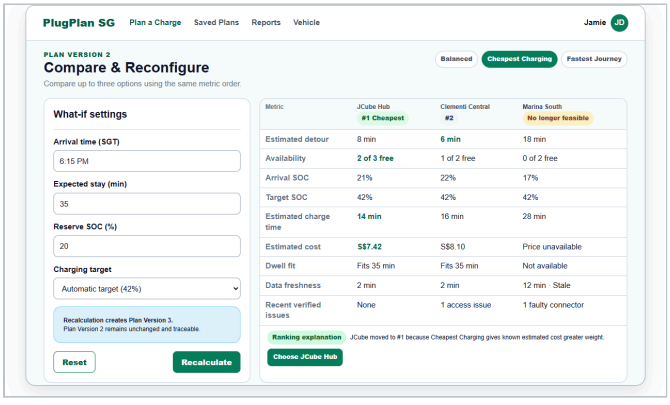
D-01 Vehicle Profile



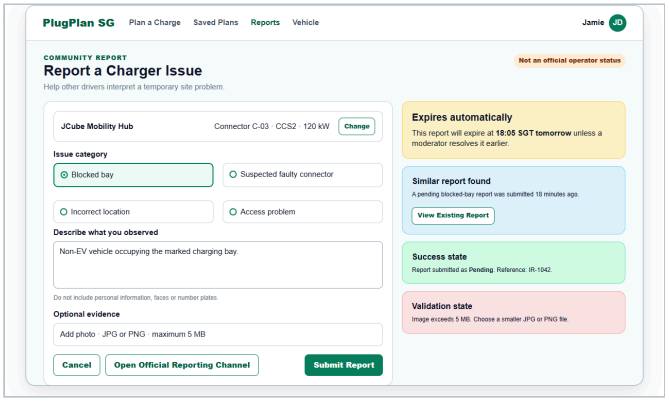
D-02 Plan a Charging Stop



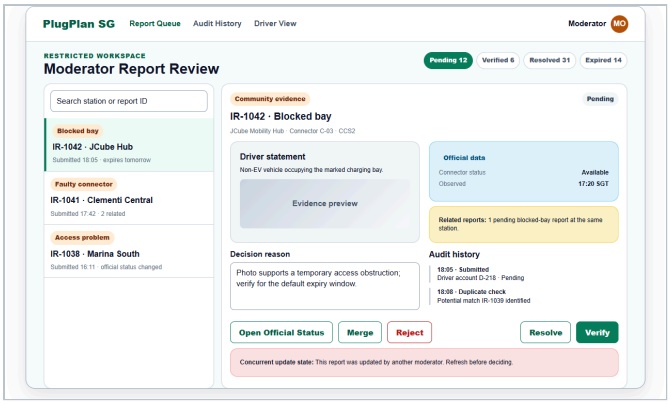
D-03 Recommendations



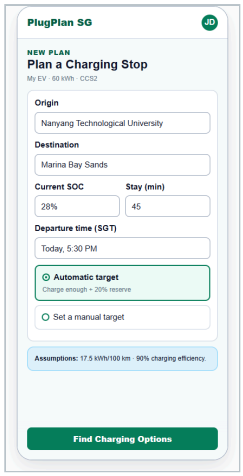
D-04 Compare and Reconfigure



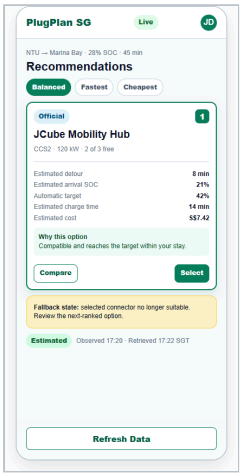
D-05 Report an Issue



D-06 Moderator Review



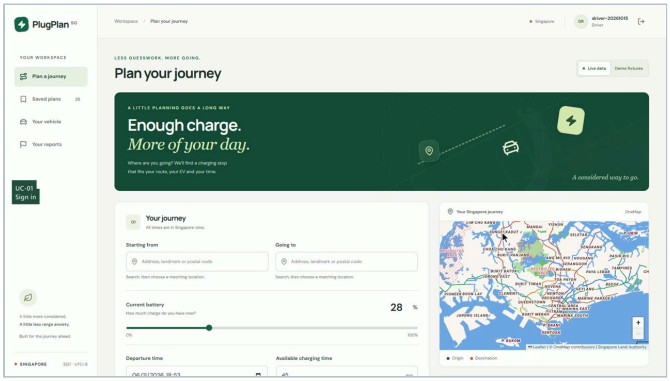
M-01 Plan (mobile)



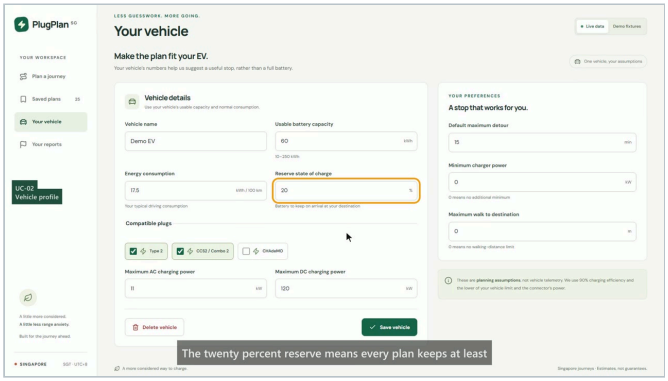
M-02 Recommendations (mobile)

B.3 Implemented Screens

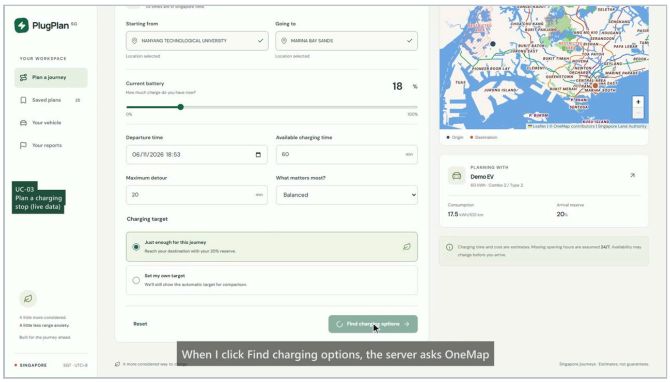
Frames captured from the submitted demo video of the final application. The caption bars and use case tags are part of the video.



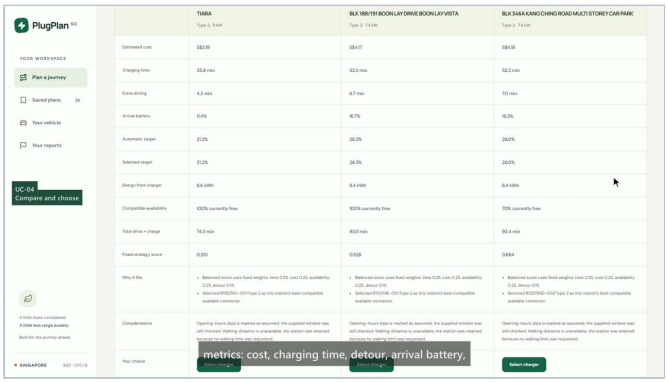
D-01 Vehicle profile



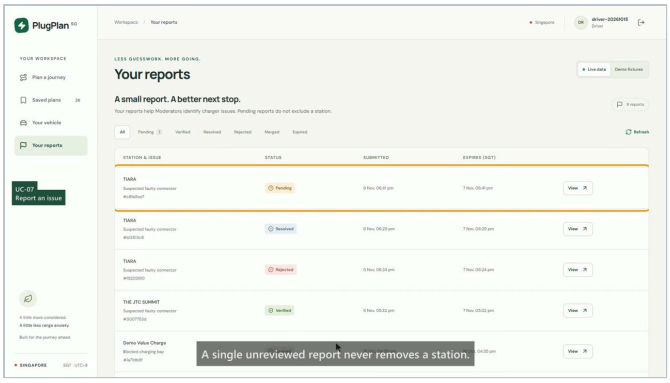
D-02 Plan input



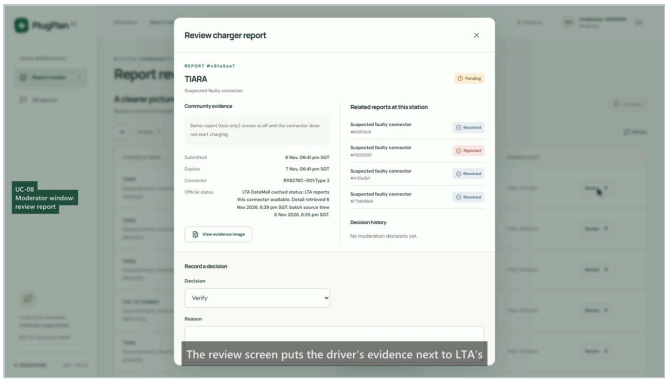
D-03 Recommendations with map



D-04 Side-by-side comparison



Reports list



D-06 Moderator report review

B.4 Design Models

Class diagrams and the UI state machine are in Class Diagrams.pdf and Dialog Map.pdf in this folder.